Serhan Ensar Büdün

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Education

Ege University İzmir, Turkey
B.Sc. in Computer Engineering (Active) *Aug. 2024 – May 2029*

Languages

Turkish: Native | **English**: Professional Working Proficiency (B2)

Technical Skills

Programming Languages: Python (Advanced), C++ (Intermediate), Kotlin (Intermediate), TypeScript (Basic)
Frameworks & Libraries: React, Electron, Vite, FastAPI, Flask, OpenCV, CustomTkinter, Tkinter, Jetpack Compose, Jinja2
AI & Vision: YOLOv8/v11 (OBB), SAHI, ByteTrack, Blender Python API (bpy), Cycles Renderer, PyInstaller
Embedded & IoT: Arduino, ESP32, Deneyap Kart, Raspberry Pi, Nextion HMI, Wear OS, RFID, IMU, STM32, LoRa
Developer Tools: Linux, Git/GitHub, Docker, VS Code, Android Studio, Roboflow
3D Design & Modeling: Fusion 360 (Advanced), SolidWorks (Intermediate), Blender (Intermediate)

Projects

LAÇİN — Aerial AI Object Detection System | *TEKNOFEST 2026 – AI in Aviation* 2025 – 2026

- Serving as **Team Captain** of a 5-member team; leading technical coordination, system architecture decisions, and server communication layer (JSON API).
- **Object Detection (personal):** Built the team's YOLOv11 training pipeline on a 116K-image dataset (VisDrone, UAVDT, Stanford Drone, DOTAv2, custom UAP/UAI labels); initial mAP@50 = 0.655. Developed a custom augmentation pipeline applying Gaussian blur and synthetic dead pixels to simulate real-field sensor failures.
- **Data Augmentation & Label Visualizer GUI (personal):** CustomTkinter + OpenCV desktop tool compositing UAP/UAI objects onto backgrounds with configurable blur and automatic YOLO label updates. Integrated label visualizer for side-by-side bounding-box verification; background threading prevents UI freezes. Packaged as standalone `.exe/.app` via PyInstaller.
- **JSONtoYOLO Converter (personal):** Tkinter desktop tool converting Supervisely-format (JSON + images) datasets to YOLOv11 structure (`images/`, `labels/`, `classes.txt`, `dataset.yaml`) with dark-themed GUI, class filter, live log stream, and background-thread processing.

TUYGUN — Aerial AI System | *TEKNOFEST 2026 – AI in Aviation* 2025 – 2026

- Serving as **Team Captain** of a 4-member team competing independently in the same category as LAÇİN; responsible for team organization, task distribution, milestone planning, and cross-team technical coordination.

EGE ODBARS — Autonomous UGV | *TEKNOFEST 2026 – Unmanned Ground Vehicle* 2025 – 2026

- Contributing as **Embedded Software & Computer Vision Member** in a 13-member multidisciplinary team building a 6x4 Rocker-Bogie UGV.
- **ODBARS NEXUS — Ground Control Station (personal):** Built a military-aesthetic tactical HUD with React, Vite, and Electron featuring real-time telemetry (speed, battery, pitch/roll), simultaneous 3-camera MJPEG streaming, 7-mission task management, and keyboard-driven autonomous/manual mode switching with dynamic targeting camera layout.
- **Synthetic Dataset Generation System (personal):** Dual-pipeline data factory in a single Tkinter GUI. *3D Pipeline:* Blender headless (`bpy`, `--background`); procedural 40x40m terrain (Polyhaven PBR + Displace modifier), fully code-generated 3D objects (traffic signs, STOP boards, shooting targets), randomized camera/lighting, automatic YOLO labeling via **3D→2D camera-matrix projection** (world → NDC → normalized pixel); Cycles GPU, 64 samples, 1920x1080. *2D Pipeline:* OpenCV + PIL compositing with perspective warp and Gaussian blur. Includes label visualizer and automated 80/20 train/val split.
- **Computer Vision (personal):** YOLOv8 object detection (traffic signs, cones, obstacles) integrated with **ByteTrack** multi-object tracking and a custom **Global Motion Compensation (GMC)** layer isolating true object motion from vehicle ego-motion drift via background optical flow subtraction.

EGENODE — Dynamic Logistics Mobile Robot | *TEKNOFEST 2026 – Robolig* 2025 – 2026

- Contributing as **Embedded Software & Hardware-Software Optimization Lead** in a 5-member team.
- Responsible for firmware on **Deneyap Kart** (ESP32, 240 MHz dual-core): multitasking for simultaneous autonomous delivery alignment and manual RC control via **iBUS protocol** over 2.4 GHz.
- System integrates RFID parcel verification (SPI), LSM6DSM 6-axis IMU for zipline stabilization, 4-DOF robotic arm (6 servo motors, cage end-effector), and H-bridge DC motor drivers. Mission flow: RFID pickup → safe-carry mode → autonomous delivery alignment → zipline return.

HomeAgent / AgentJee — AI-Powered Smart Home Ecosystem | *Python, FastAPI, Gemini AI, Kotlin, ESP32* 2025

- Built a central smart home server on **Raspberry Pi** serving as the hub for Android (HomeAgentMobile) and Wear OS (AgentJee) clients.
- **Backend (HomeAgent):** FastAPI async REST with JWT session auth and Jinja2 web UI; 20+ endpoints covering real-time CPU/RAM/disk/temp monitoring, full file manager (upload, download, move, copy, rename, search, preview, trash), Docker container management, and network status. **Google Gemini AI** integration enables voice/text command understanding for autonomous home control decisions.
- **Wear OS App (AgentJee):** Native Kotlin/Jetpack Compose interface for Samsung Galaxy Watch 4 Classic with bezel navigation, 3s auto-refresh dashboard, and system reboot/shutdown controls via OkHttp.
- **ESP32 Hardware Panel:** Custom C++ (Arduino) touchscreen menu on ESP32 + ILI9341 TFT (TFT_eSPI) connecting over WiFi for physical server navigation.
- **Telegram Bot:** Remote management via /status, /reboot, /shutdown, /ip, /disk commands with inline keyboard menu.

Smart Telemetry System — WearOS → ESP32 → Nextion HMI | *Embedded / Health IoT* 2025

- Real-time telemetry pipeline streaming HR, SpO2, accelerometer, and gyroscope data (1 Hz) from Wear OS via HTTP/JSON to ESP32; simultaneously drives a live browser dashboard (CSV log download) and a **Nextion TFT display** with visual gauges and IMU waveform graphs.

TÜBİTAK Electric Motor Innovation | *R&D, Electromechanics* 2019

- Co-designed a 2-person R&D project analyzing electric motor efficiency and dynamic force development; conducted comparative experiments to optimize foundational motor mechanics.

Extracurricular Activities

IEEE Ege University Student Branch | *Computer Society Chairperson* Sept. 2024 – Present

- Led the Computer Society committee, managing strategic planning and supervision of **6 technical sub-teams**.
- Designed and instructed a comprehensive **5-week Arduino & Circuit Design training course**, mentoring peers in C++ and embedded systems.
- Coordinated major organizational events and represented the university at national summits including **CSCAMP** and the **Ege Regional Meeting**.

Competitions & Conferences | *Hackathons, Summits* 2024 – 2025

- Competed in **IEEEXtreme**, a global 24-hour algorithmic programming hackathon; participated in **Google DevFest** to engage with industry professionals and emerging technologies.